CONTACT US

法奥意威(苏州)机器人系统有限公司
FAIR Innovation (Suzhou) Robot System Co.,Ltd.

总部：江苏省苏州市高新区竹园路209号
生产制造一厂：山东省淄博市高新区尊贤路5888号
生产制造二厂：江苏省苏州市吴中区紫金路36号

R&D Center : No.209 Zhuyuan Road,New District,Suzhou,Jiangsu,PR.China
Manufacture Base 1 : No.5888 Zunxian Road,New District,Zibo,Shandong,PR.China
Manufacture Base 2 : No.36 Zijin Road,Wuzhong District,Suzhou,Jiangsu,PR,China

400-811-0929 www.frtech.cc Marketing@frtech.fr

企业官网 微信公众号 微信视频号 资料下载

FAIRINO
—法奥机器人—

PRODUCT MANUAL

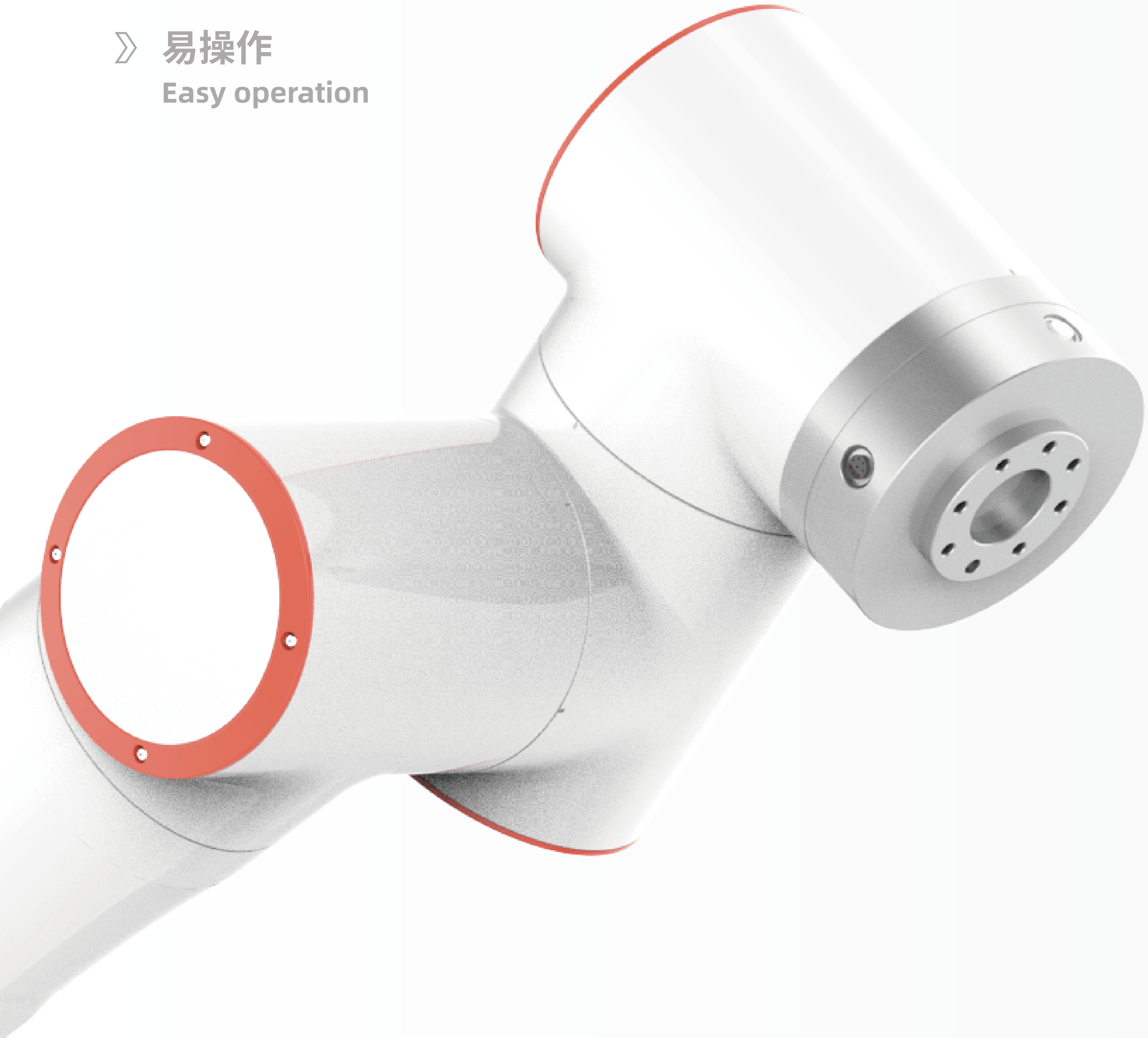
产品手册

》 **模块化**
Modularization

》 **快部署**
Quick deployment

》 **易操作**
Easy operation

FAIRINO
ROBOT



PRODUCT VISION
产品愿景

协作机器人将延伸您的时间和空间，解放繁杂的低效重复，让您的双手和大脑拥抱更广袤的天际。未来，您将见证**法奥协作机器人**与众协同。
Collaborative robots will extend your time and space, liberate complex and inefficient repetition, and let you embrace a wider horizon.In the future, You will see **FAIRINO ROBOTS**.

随着它的出现，不仅提高了人机协作的效率，也加快了更多企业自动化进程，对许多制造商来说，它还节省了空间，降低了部署机器人产线的成本。
With its emergence, it not only improves the efficiency of human-machine collaboration, but also speeds up the automation process for more enterprises and frees floor space and lowers the cost of implementing robots for manufacturers.

PRODUCT DISPLAY

产品展示

FR3 FR3-C FR3-WMS FR3-WML FR5 FR10 FR16 FR20 FR30

FR3及系列产品

FAIRINO FR SERIES

系列排布

根据不同的负载能力和参数范围，法奥协作机器人产品FR系列拥有九个型号：FR3（可选镜像版本）、FR3-C、FR3-WMS、FR3-WML、FR5、FR10、FR16、FR20和FR30。法奥通过国际权威机构已获取更全产品认证，为伙伴以及客户提供更优品质保障。

According to different payload and parameter, FAIRINO collaborative robots FR series are divided into nine models:FR3(Optional Mirror Version), FR3-C, FR3-WMS,FR3-WML, FR5, FR10, FR16, FR20 and FR30. To provide partners&-customers with better quality assurance,FAIRINO has obtained a more comprehensive range of certificates through international certification organizations.

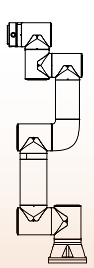
法奥协作机器人FR系列产品生产通过了ISO 9001质量管理体系认证
Quality Management System: ISO 9001

产品认证：CR, CE, KCs, NRTL, RoHS 2.0, NSF, SEMI, IP65
Product Certification: CR, CE, KCs, NRTL, RoHS 2.0, NSF, SEMI, IP65

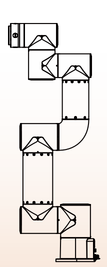
ISO功能安全认证：ISO 10218, ISO 13849, ISO 15066
ISO Functional Safety Certification: ISO10218, ISO13849, ISO15066

智能人机系统解决方案
Intelligent human-robot cooperation system solutions

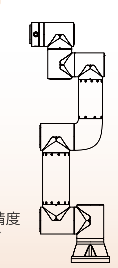
FR3
有效负载 Payload
额定:3kg
最大:5kg
工作半径 Reach
622mm
重复定位精度 Repeatability
±0.02mm



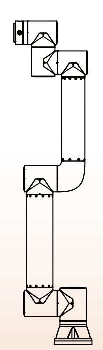
FR3-C
有效负载 Payload
额定:3kg
瞬时:5kg
工作半径 Reach
622mm
重复定位精度 Repeatability
±0.05mm



FR3-WMS
有效负载 Payload
3kg
工作半径 Reach
622mm
重复定位精度 Repeatability
±0.02mm

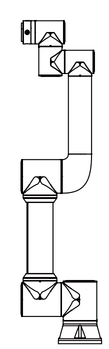


FR3-WML
有效负载 Payload
3kg
工作半径 Reach
922mm
重复定位精度 Repeatability
±0.05mm

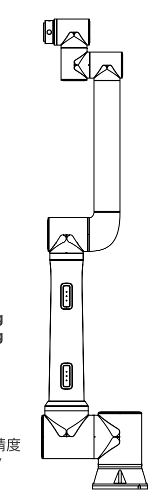


FR3及系列产品

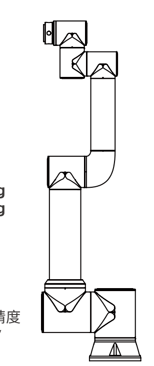
FR5
有效负载 Payload
额定:5kg
最大:7kg
工作半径 Reach
922mm
重复定位精度 Repeatability
±0.02mm



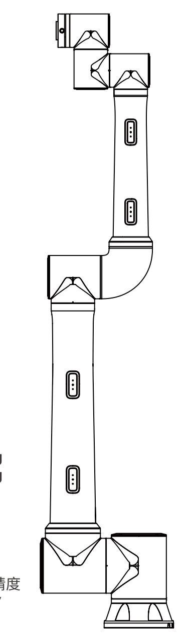
FR10
有效负载 Payload
额定:10kg
最大:14kg
工作半径 Reach
1400mm
重复定位精度 Repeatability
±0.05mm



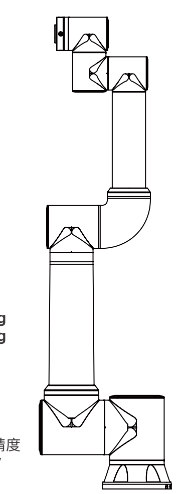
FR16
有效负载 Payload
额定:16kg
最大:20kg
工作半径 Reach
1034mm
重复定位精度 Repeatability
±0.03mm



FR20
有效负载 Payload
额定:20kg
最大:25kg
工作半径 Reach
1854mm
重复定位精度 Repeatability
±0.1mm



FR30
有效负载 Payload
额定:30kg
最大:35kg
工作半径 Reach
1403mm
重复定位精度 Repeatability
±0.1mm



ROBOT ARM TECHNICAL SPECIFICATION

机械臂规格参数

>> FR3及系列产品 <<

	FR3	FR3-C	FR3-WMS	FR3-WML	FR5	FR10	FR16	FR20	FR30
有效负载 (Payload)	3kg (最大:5kg)	3kg (瞬时:5kg)	3kg	3kg	5kg (最大:7kg)	10kg (最大:14kg)	16kg (最大:20kg)	20kg (最大:25kg)	30kg (最大:35kg)
工作半径 (Reach)	622mm	622mm	622mm	922mm	922mm	1400mm	1034mm	1854 mm	1403 mm
自由度 (Degrees of freedom)	6个旋转关节	6个旋转关节	6个旋转关节	6个旋转关节	6个旋转关节	6个旋转关节	6个旋转关节	6个旋转关节	6个旋转关节
人机交互 (HMI)	10.1 英寸示教器或移动终端 Web App		10.1 英寸示教器或移动终端Web App				10.1 英寸示教器或移动终端		
双臂机器人应用 (Dual arm robotics applications)	有镜像版本，可组建双臂机器人								
符合ISO 9283的位姿可重复性 (Pose repeatability per ISO 9283)	±0.02mm	±0.05mm	±0.02mm	±0.05mm	±0.02mm	±0.05mm	±0.03mm	±0.1mm	±0.1mm

	FR3		FR3-C		FR3-WMS		FR3-WML		FR5		FR10		FR16		FR20		FR30	
轴移动(Axis movement)	工作范围	最大速度	工作范围	最大速度	工作范围	最大速度	工作范围	最大速度	工作范围	最大速度	工作范围	最大速度	工作范围	最大速度	工作范围	最大速度	工作范围	最大速度
基座(Base)	±175°	±180°/s	±175°	±150°/s	±175°	±150°/s	±175°	±150°/s	±175°	±180°/s	±175°	±120°/s	±175°	±120°/s	±175°	±120°/s	±175°	±120°/s
肩部 (Shoulder)	+ 85°/- 265°(双臂:- 85°/ + 265°)	±180°/s	+ 85°/ - 265°	±150°/s	+ 85°/ - 265°	±150°/s	+ 85°/ - 265°	±150°/s	+ 85°/ - 265°	±180°/s	+ 85°/ - 265°	±120°/s	+ 85°/ - 265°	±120°/s	+ 85°/ - 265°	±120°/s	+ 85°/ - 265°	±120°/s
肘部 (Elbow)	±150°	±180°/s	±150°	±150°/s	±150°	±150°/s	±150°	±150°/s	±160°	±180°/s	±160°	±180°/s	±160°	±180°/s	±160°	±120°/s	±160°	±120°/s
腕部 1 (Wrist 1)	+ 85°/- 265°(双臂:- 85°/ + 265°)	±180°/s	+ 85°/ - 265°	±180°/s	+ 85°/ - 265°	±180°/s	+ 85°/ - 265°	±180°/s	+ 85°/ - 265°	±180°/s	+ 85°/ - 265°	±180°/s	+ 85°/ - 265°	±180°/s	+ 85°/ - 265°	±180°/s	+ 85°/ - 265°	±180°/s
腕部 2 (Wrist 2)	±175°	±180°/s	0°~355°	±180°/s	±175°	±180°/s	±175°	±180°/s	±175°	±180°/s	±175°	±180°/s	±175°	±180°/s	±175°	±180°/s	±175°	±180°/s
腕部 3 (Wrist 3)	±175°（选配 ±360°）	±180°/s	±175°	±180°/s	±360°	±180°/s	±360°	±180°/s	±175° (选配 ±360°)	±180°/s	±175°(选配 ±360°)	±180°/s	±175° (选配 ±360°)	±180°/s	±175°	±180°/s	±175°	±180°/s

	FR3	FR3-C	FR3-WMS	FR3-WML	FR5	FR10	FR16	FR20	FR30
典型 TCP 速度(Typical TCP speed)	1m/s	1m/s	1m/s	1m/s	1m/s	1.5m/s	1m/s	2m/s	2m/s
防护等级(IP classification)	IP54 (可选 IP65)	IP54	IP54 (可选 IP65)	IP54 (可选 IP65)	IP54 (可选 IP65)	IP54 (可选 IP65)	IP54 (可选 IP65)	IP54 (可选 IP65)	IP54 (可选 IP65)
噪音(Noise)	<65dB	<65dB	<65dB	<65dB	<65dB	<65dB	<65dB	<70dB	<70dB
安装方向(Robot mounting)	任何方向	任何方向	任何方向	任何方向	任何方向	任何方向	任何方向	任何方向	任何方向
I/O端口(I/O Ports)	数字输入(DI) 2 数字输出(DO) 2	数字输入(DI) 2 数字输出(DO) 2	数字输入(DI) 2 数字输出(DO) 2	数字输入(DI) 2 数字输出 (DO) 2	数字输入(DI) 2 数字输出(DO) 2	数字输入(DI) 2 数字输出(DO) 2	数字输入(DI) 2 数字输出(DO) 2	数字输入(DI) 2 数字输出(DO) 2	数字输入(DI) 2 数字输出(DO) 2
	模拟输入(AI) 1 模拟输出(AO) 1	模拟输入(AI) 1 模拟输出(AO) 1	模拟输入(AI) 1 模拟输出(AO) 1	模拟输入(AI) 1 模拟输出 (AO) 1	模拟输入(AI) 1 模拟输出(AO) 1	模拟输入(AI) 1 模拟输出(AO) 1	模拟输入(AI) 1 模拟输出(AO) 1	模拟输入(AI) 1 模拟输出(AO) 1	模拟输入(AI) 1 模拟输出(AO) 1
工具I/O电源 (Tool I/O power supply)	24V/1.5A	24V/1.5A	24V/1.5A	24V/1.5A	24V/1.5A	24V/1.5A	24V/1.5A	24V/1.5A	24V/1.5A
底座直径(Footprint)	128mm	125mm	128mm	128mm	149mm	190mm	190mm	240mm	240mm
整机重量(Weight)	≈15kg	≈10kg	≈10.5kg	≈11.25kg	≈22kg	≈40kg	≈40kg	≈85kg	≈85kg
工作温度(Operating temperature)	0-45℃	0-45℃	0-45℃	0-45℃	0-45℃	0-45℃	0-45℃	0-45℃	0-45℃
工作湿度(Operating humidity)	90%RH(non-condensing)	90%RH(non-condensing)	90%RH(non-condensing)	90%RH(non-condensing)	90%RH(non-condensing)	90%RH(non-condensing)	90%RH(non-condensing)	90%RH(non-condensing)	90%RH(non-condensing)
设备材料(Materials)	铝、钢	铝、钢	铝、钢	铝、钢	铝、钢	铝、钢	铝、钢	铝、钢	铝、钢
黑色选配（Black Optional)	否	否	否	否	是	是	否	否	否

■ 典型功率测试负载设置，根据机器型号设置不同的负载,负载配置参数如下：
Typical power test payload settings, different loads are set according to robot models, payload configuration parameters are as follows :

	FR3	FR3-C	FR3-WMS	FR3-WML	FR5	FR10	FR16	FR20	FR30
	FR3负载设置:3kg，Z轴:18mm FR3 payload setting: 3kg, Z-axis: 18mm	FR3C负载设置:3kg，Z轴:18mm FR3C payload setting: 3kg, Z-axis: 18mm	FR3WMS负载设置:3kg，Z轴:18mm FR3WMS payload setting: 3kg, Z-axis: 18mm	FR3WML负载设置:3kg，Z轴:18mm FR3WML payload setting: 3kg, Z-axis: 18mm	FR5负载设置:5kg，Z轴:30mm FR5 payload setting: 5kg, Z-axis: 30mm	FR10负载设置:10kg，Z轴:60mm FR10 payload setting: 10kg, Z-axis: 60mm	FR16负载设置:16kg，Z轴:96mm FR16 payload setting: 16kg, Z-axis: 96mm	FR20负载设置:20kg，Z轴:120mm FR20 payload setting: 20kg, Z-axis: 120mm	FR30负载设置:30kg，Z轴:200mm FR30 payload setting: 30kg, Z-axis: 200mm

选用老化测试程序，机器人总电源接入功率计，机器人设置自动模式，全局速度设置为100，点击运行,运行两个周期无异常则开始持续测试，持续24小时。24小时后分别统计功率计的峰值和平均功率，依次对各个型号进行统计：
Select aging test program, connect robot's total power to power meter, set robot to automatic mode, set global speed to 100, click run, if there are no abnormalities after running two cycles, start continuous testing for 24 hours. After 24 hours, respectively, record the peak and average power of the power meter, and then statistically analyze each model :

	FR3	FR3-C	FR3-WMS	FR3-WML	FR5	FR10	FR16	FR20	FR30
典型平均功率 (Typical average power)	220W	200W	90W	140W	260W	300W	310W	620W	600W
典型峰值功率 (Typical peak power)	280W	230W	130W	240W	310W	500W	410W	810W	910W

CONTROLLER TECHNICAL SPECIFICATION

控制箱规格参数

	直流mini控制箱2kW	直流控制箱5kW	交流mini控制箱2kW	交流控制箱5kW
设备特性 Features				
防护等级(IP classification)	IP54	IP54	IP54	IP54
工作温度(Operating temperature)	0-45℃	0-45℃	0-45℃	0- 45℃
工作湿度(Operating humidity)	90%RH(non-condensing)	90%RH(non-condensing)	90%RH(non-condensing)	90%RH(non-condensing)
I/O端口(I/O Ports)	数字输入(DI) 16 数字输出(DO) 16	数字输入(DI) 16 数字输出(DO) 16	数字输入(DI) 16 数字输出(DO) 16	数字输入 (DI) 16 数字输出(DO) 16
	模拟输入(AI) 2 模拟输出(AO) 2	模拟输入(AI) 2 模拟输出(AO) 2	模拟输入(AI) 2 模拟输出(AO) 2	模拟输入(AI) 2 模拟输出(AO) 2
	高速脉冲输入(High speed pulse input) 2	高速脉冲输入(High speed pulse input) 2	高速脉冲输入(High speed pulse input) 2	高速脉冲输入 (High speed pulse input) 2
I/O电源(I/O power supply)	24V/1.5A	24V/1.5A	24V/1.5A	24V/1.5A
标配通讯(Standard communication)	I/O、TCP/IP、Modbus_TCP/RTU	I/O、TCP/IP、Modbus_TCP/RTU	I/O、TCP/IP、Modbus_TCP/RTU	I/O、TCP/IP、Modbus_TCP/RTU
可选通讯(Optional communication)	CC-Link IE Field Basic、Profinet、Ethernet/IP、EtherCAT	CC-Link IE Field Basic、Profinet、Ethernet/IP、EtherCAT	CC-Link IE Field Basic、Profinet、Ethernet/IP、EtherCAT	CC-Link IE Field Basic、Profinet、Ethernet/IP、EtherCAT
通讯板卡选配 (Communication Board Optional Configuration)	MiniPCI Express- 实时以太网 PC板卡	MiniPCI Express- 实时以太网 PC板卡	MiniPCI Express- 实时以太网 PC板卡	MiniPCI Express- 实时以太网 PC板卡
软件开发包 (Software development kit)	C#/C++/Python/ROS/ROS2	C#/C++/Python/ROS/ROS2	C#/C++/Python/ROS/ROS2	C#/C++/Python/ROS/ROS2
物理性能 Physical				
尺寸参数(L*W*H)	245*180*44.5mm(不含凸出物)	245*180*89mm(不含凸出物)	245*180*44.5mm (不含凸出物)	245 *180*89mm (不含凸出物)
设备重量(Weight)	2.1kg (不含线重量)	2.957 kg (不含线重量)	2.5kg (不含线重量)	3.6 kg (不含线重量)
设备材料(Materials)	镀锌板 Galvanized plate	镀锌板 Galvanized plate	镀锌板 Galvanized plate	镀锌板 Galvanized plate
供电电源(Power supply)	30-60VDC	30-60VDC	220VAC / 10A / 单相 / 50Hz (国内版) 100-240VAC / 10A / 单相 / 50-60Hz (海外版)	100-240VAC / 16A / 单相 / 50-60Hz
输出功率(Output power)	48VDC / 42Amax	48VDC / 104Amax	48VDC / 42Amax	48VDC / 104Amax
适配机器人 Applicable Robot	FR3,FR3-WMS,FR3-WML,FR5,FR10,FR16	FR20/FR30	FR3,FR3-WMS,FR3-WML,FR5,FR10,FR16	FR20/FR30

SAFETY BOX
按钮盒

防护等级(IP classification)	IP54
按钮功能(Button function)	手动/自动、拖动、点记录、是否配合按钮盒、开始/停止、关机 Manual/Auto, Drag, Point Record, Match or Not with Safety Button Box, Start/Stop, Shutdown
协议类型(Communication)	TCP/IP
网络传输速率(Network transfer rate)	100M
供电(Power over ethernet)	标准POE Standard POE
尺寸参数(L*W*H)	136*60*66mm (不含凸出物) (No protrusions)
设备重量(Weight)	490g (带线重量) (Cable weight included)
设备材料(Materials)	ABS
线缆长度(Cable length)	5m
按键次数(Number of keys)	≥20W 次

人机交互工具，可实现基础交互功能。通过RJ45接口可与电脑、平板等设备连接，直接登录到WebAPP。

Human-cobot interaction tools for basic interaction functions. It can be linked with computers, tablets and other devices through the RJ45 interface, and directly log in to the Web App teaching interface.

TEACH PENDANT [选配 Optional]
示教器

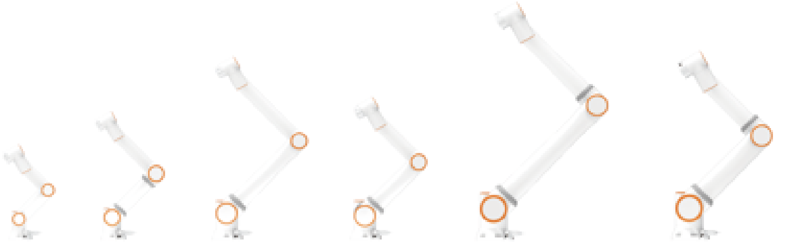
防护等级(IP classification)	IP54
工作湿度(Operating humidity)	90%RH(non-condensing)
显示分辨率(Display resolution)	1280 x 800 pixels
尺寸参数(L*W*H)	268*210*88mm
设备重量(Weight)	1.6kg
设备材料(Materials)	ABS、PP
线缆长度(Cable length)	5m

示教器、电脑、平板或手机与WebAPP系统连接，从而实现对协作机器人的操控。

The teach pendant, computer, tablet or mobile phone is connected to the WebAPP system to realize the operation of the collabroative robot.

EXPLOSION-PROOF CABINET [选配 Optional]
防爆柜

防爆柜(整体防爆)
Explosion-Proof Cabinets (Integral Explosion-Proof)



机器人(全系列防爆)
Robot (full range explosion-proof)

具有良好的技术性能。机柜应具有抗振动、抗冲击、耐腐蚀、防尘、防水、防辐射等性能，以便保证设备稳定可靠地工作。良好的使用性和安全防护设施，便于操作、安装和维修，并能保证操作者安全。

It has good technical performance. The cabinet must be resistant to vibration, impact, corrosion, dust, water, and radiation to ensure stable and reliable operation of devices. Good usability and safety protection facilities, easy operation, installation and maintenance, and to ensure the safety of the operator.

防护等级(IP classification)	IP65
工作环境温度(Operating Environment Temperature)	0°C-45°C
最低正压(Minimum Positive Pressure)	100pa
最高正压(Maximum Positive Pressure)	1000pa
额定输入电压(Rated Input Voltage)	100–240Vac
最大泄漏量(Maximum Leakage)	15L/Min
尺寸参数(L*W*H)	682*500*1100(含报警灯1286)
设备重量(Weight)	75kg
设备材料(Materials)	碳钢+不锈钢
柜内容积(Cabinet Volume)	126L
换气流量(Ventilation Flow Rate)	120L/min
换气时间(Ventilation Time)	14min
保护气体(Protective Gas)	空气(AIR)
防爆柜证书(Explosion-proof Cabinet Certificate)	CE22.7131

INDUSTRY APPLICATIONS

上下料解决方案

Pick And Place Solution

上下料机器人能够提高生产效率、质量和安全性，降低劳动强度，并提高灵活适应性，为企业带来更高的效益和竞争优势。

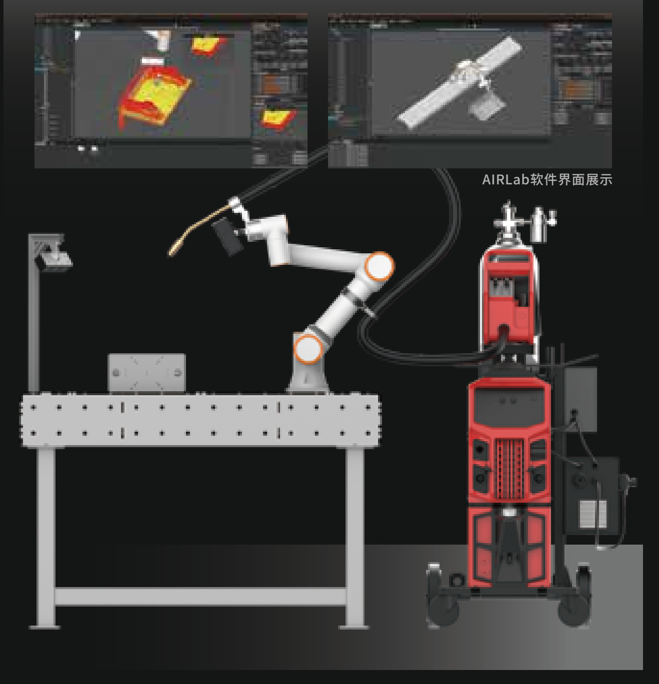
Material handling robots can improve production efficiency, quality, and safety, reduce labor intensity, and provide flexibility and adaptability, bringing higher benefits and competitive advantages to businesses.

3D Vision Programming-free Welding Solution

3D视觉免编程焊接解决方案

3D视觉免编程焊接解决方案，集成法奥机器人、AIRLab、3D相机，搭载丰富的焊接工艺包，通过3D相机扫描结合点云算法进行工件定位、焊缝识别，基于AI深度学习进行动态路径规划及焊接工艺匹配，实现智能焊接。

3D Vision Programming-Free Welding Solution integrates FAIRINO robot, AIRLab, and 3D camera, equipped with a comprehensive welding process package. By leveraging 3D camera scanning and point cloud algorithms for workpiece positioning and weld seam recognition, it utilizes AI deep learning for dynamic path planning and welding process matching, enabling intelligent welding.

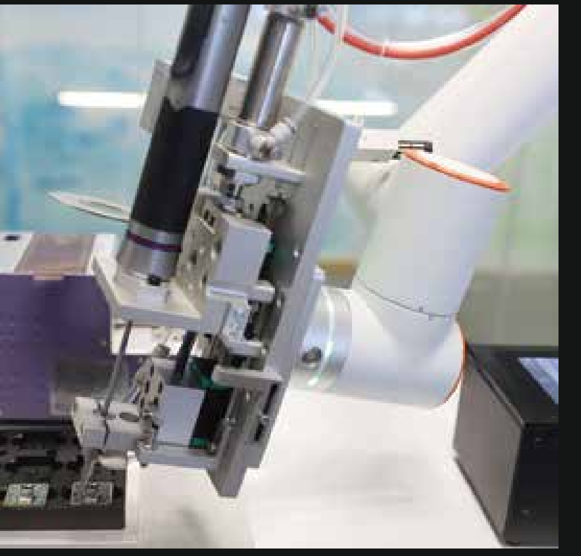


Screw Tightening Robot

螺丝锁付机器人

搭配末端智能拧紧装置，实现了扭矩可调、可控、可编辑，适合各种场景下的螺丝锁付，能够稳定、高效、准确完成生产过程，极大减少工人重复性劳动，支持数据溯源。

Combined with the end intelligent tightening device, it achieves adjustable, controllable, and programmable torque, making it suitable for screw locking in various scenarios. It can stably, efficiently, and accurately complete the production process, greatly reducing repetitive labor for workers and supporting data traceability.



涂胶解决方案

Glue Dispending Solution

搭配末端智能出胶装置，精细化作业，适合多场景下的精细化涂胶、点胶工作，能够稳定、高效、准确完成涂胶过程，保证涂胶效果，极大减少工人重复性劳动，保护工人健康。

Paired with an intelligent dispensing device at the end effector, it enables precise operations and is suitable for precise gluing and dispensing tasks in various scenarios. It can achieve stable, efficient, and accurate adhesive application, ensuring the quality of the adhesive work. This greatly reduces repetitive labor for workers and protects their health.

传送带解决方案

Conveyor Belt Solution

- 提升工作安全性 Enhance work safety
- 实时监控与反馈 Real-time monitoring and feedback
- 降低错误率和损耗 Reduce error rate and losses
- 提高生产效率 Improve production efficiency
- 数据记录与溯源 Data recording and traceability
- 精确追踪与识别 Accurate tracking and identification

COMMERCIAL APPLICATIONS

茶饮机器人
Automated Tea Robot

节约人力成本，替代人工，提高工作效率；冲泡茶饮口感一致，改善不同人操作、不同时间点操作而带来的差异性；具有表演性，给消费者带来乐趣，而员工可以去完成更有成就感、待遇更高的工作；成本低，快速回本，经济效益好；占地面积小，坪效更高，能适应多种新颖的创业模式。

协作机器人可以应用在多类型的新零售场景，并且根据不同场景需求，进行个性化定制。

Collaborative robots can be applied in various types of new retail scenarios and can be customized according to different scenario requirements. Benefits include:

Cost-saving: They replace manual labor, reducing manpower costs while increasing work efficiency.

Consistent tea brewing: They ensure consistent taste regardless of different operators or different time points, eliminating variations caused by human factors.

Entertainment value: The robotic performance brings enjoyment to consumers, while employees can focus on more fulfilling and higher-paying jobs.

Cost-effective: They have low costs and provide a quick return on investment, resulting in good economic benefits.

Small footprint: They occupy less space, resulting in higher space utilization and adaptability to various innovative business models.

康复解决方案
Rehabilitation Solution

实现了上肢康复与下肢锻炼的一体化，通过运动轨迹的复现，降低了使用门槛。通过实时记录反馈数据，显著提升安全性能。多种模式设定，让康复治疗更有针对性，显著提升康复效率。

It has achieved integration of upper limb rehabilitation and lower limb exercise, reducing the barrier to entry through the reproduction of motion trajectories. By recording real-time feedback data, it significantly enhances safety performance. With various mode settings, it makes rehabilitation treatment more targeted, leading to a significant improvement in rehabilitation efficiency.

艾灸解决方案
Moxibustion Solution

完整复刻五大灸法，提供悬停灸、雀啄灸、回旋灸、往复灸和循经灸，降低艾灸门槛。经过最新认证，搭配末端碰撞检测、温控、红外测距，三重防护保障艾灸安全。内置吸入装置，避免艾灸过程烟尘吸入。

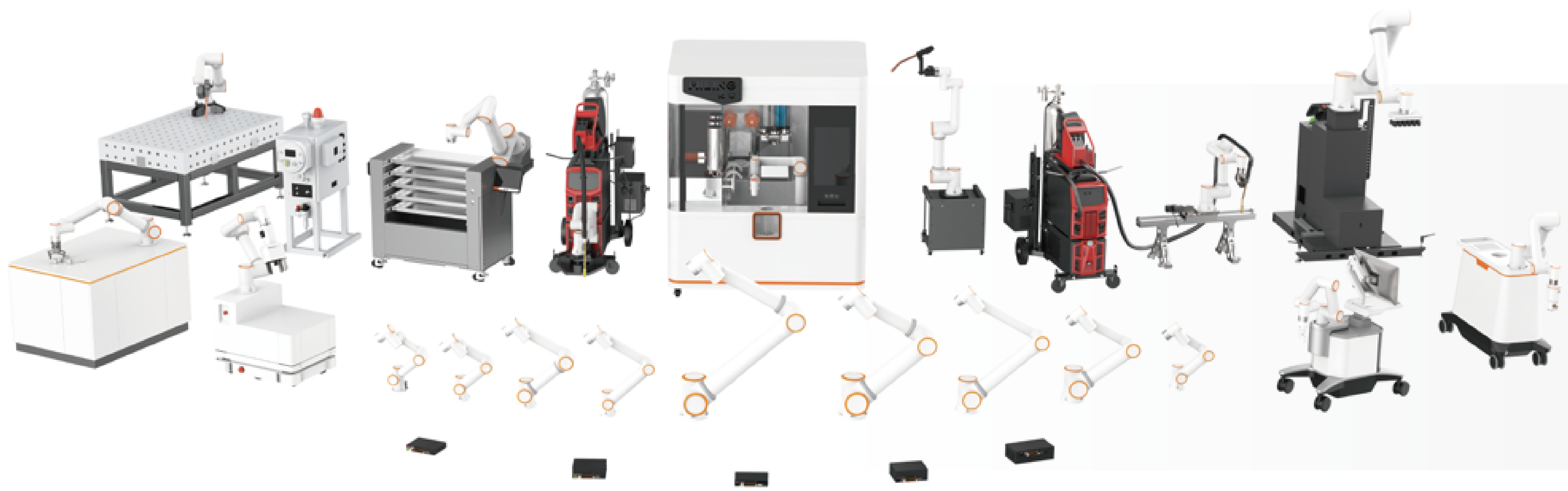
- 极致安全
- 部署灵活
- 门槛降低
- 高效艾灸

It fully replicates the five major moxibustion techniques, offering hovering moxibustion, sparrow pecking moxibustion, rotating moxibustion, reciprocating moxibustion, and meridian moxibustion, thus reducing the barrier to entry for moxibustion. With the latest certifications, it is equipped with end collision detection, temperature control, and infrared distance measurement, providing triple protection to ensure the safety of moxibustion.

It also has a built-in suction device to prevent inhalation of smoke and dust during the moxibustion process.

- Ultimate safety
- Flexible deployment
- Lower barrier to entry
- Efficient moxibustion

COMPANY PROFILE
公司简介



FAIRINO ROBOT

FAIRINO，优先实现全核心零部件自主研发的协作机器人公司。

我们关注用户体验，凭借阶梯式产品系列，融合AI、视觉、传感等最新技术，致力于为各行各业提供极致便利的深度智能系统，实现从工业到商业的普及应用。

我们为行业客户提供定制化的部件、整机及系统，开放的开发平台为我们的合作伙伴提供更多的便捷和可能。

FAIRINO始终秉持与合作伙伴、客户携手共进的理念，持续为其提供超额价值。

FAIRINO邀您一同探索，开启智能化的多元世界。

FAIRINO is the collaborative robot company who has achieved independent R&D of all core components.

We focus on user experience and are dedicated to offering the industry with artificial intelligent robot system.

We provide customized components, complete machines and systems for industry customers, the open development platform provides more convenience and possibility for our partners.

FAIRINO remains committed to the philosophy of advancing together with partners and customers, continuously delivering extraordinary value to them.

FAIRINO invites you to explore together and open up a diverse intelligent world.

智能制造时代，协作机器人如何重构价值？

· **企业增效：**降本增质，赋能员工技能提升，驱动精益生产；

· **商业升级：**优化服务体验，精准满足客户需求，打造差异化竞争力；

· **匠心传承：**标准化手作流程，平衡效率与品质，激活传统工艺新生机；

· **生活助手：**智能分担家务，定制清洁烹饪方案，让科技温暖日常。

从工业制造到商业服务，从匠人工坊到智慧家居，法奥机器人持续拓展应用边界，让未来场景触手可及。

In the era of intelligent manufacturing, how can collaborative robots redefine value?

·Business Efficiency: Reduce costs, enhance quality, empower skill upgrading, and drive lean manufacturing.

·Commercial Upgrade: Optimize service experiences, meet customer needs precisely, and build differentiated competitiveness.

· Craft Heritage: Standardize artisanal processes, balance efficiency and quality, and revitalize traditional craftsmanship.

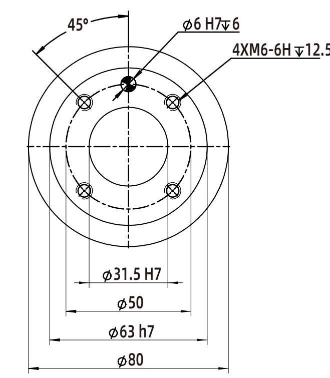
·Life Companion: Intelligently handle household chores, customize cleaning and cooking solutions, and infuse technology into daily life.

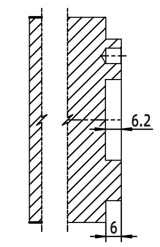
From industrial manufacturing to commercial services, from artisan workshops to smart homes, FAIRINO Robotics continues to expand the boundaries of application, making future scenarios tangible today.

DRAWINGS

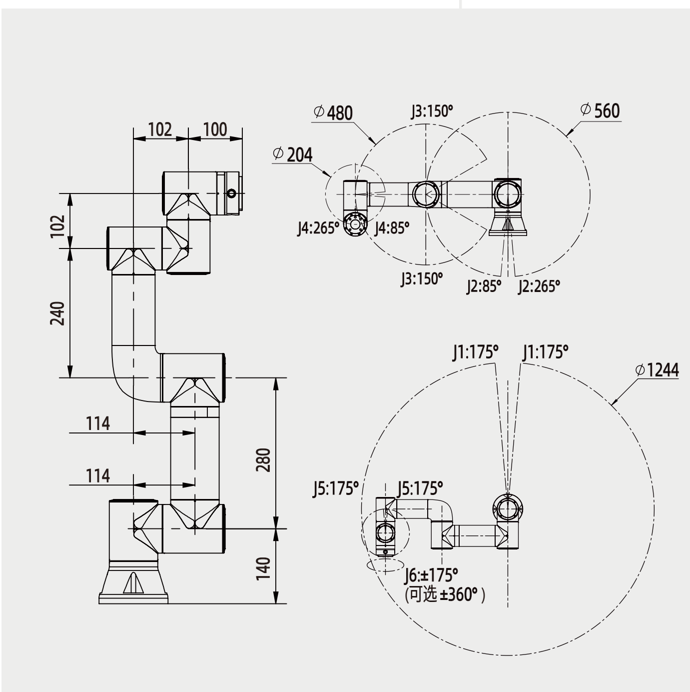
技术图纸

单位(Unit) : mm

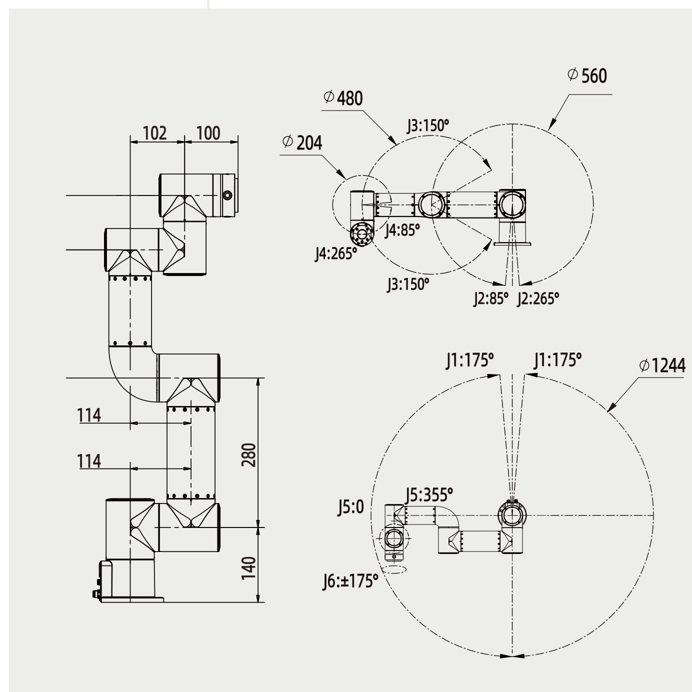




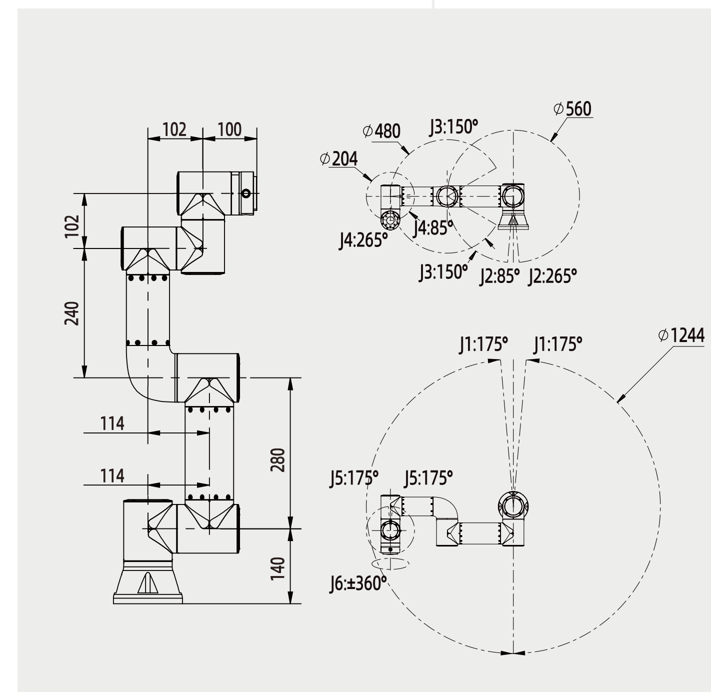
机器人末端
均采用国际标准



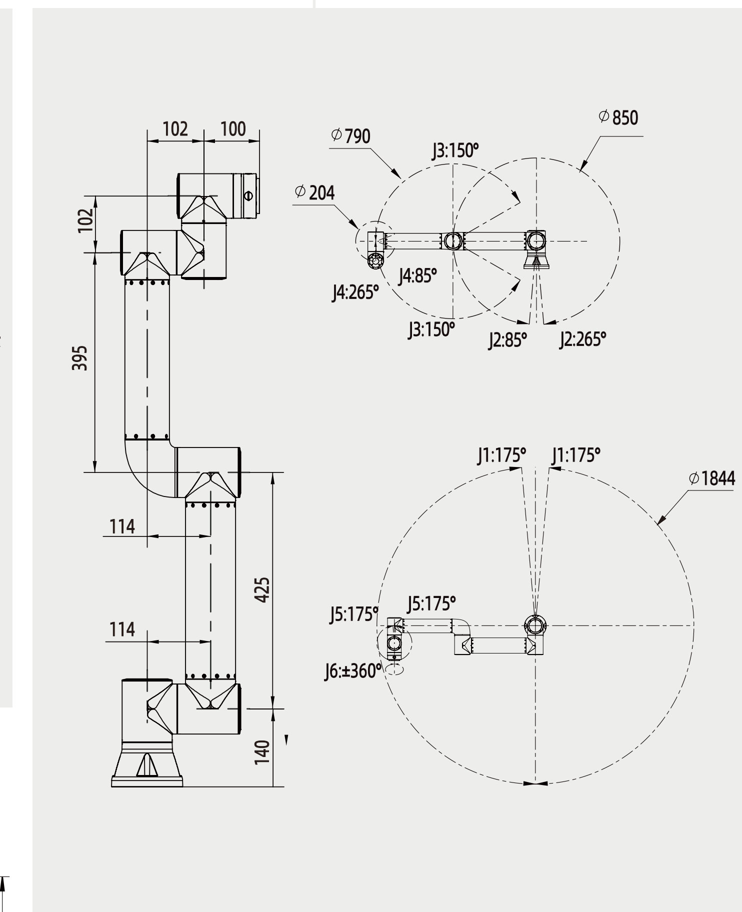
FR3



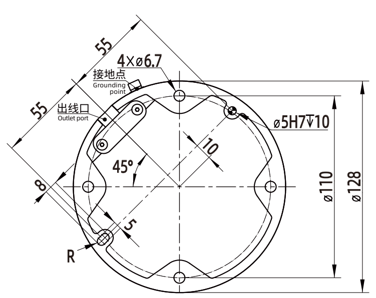
FR3-C



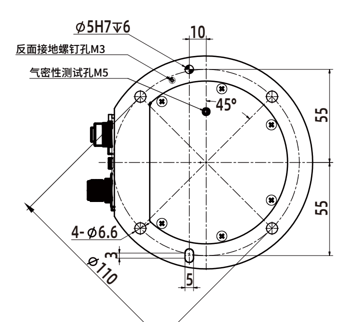
FR3-WMS



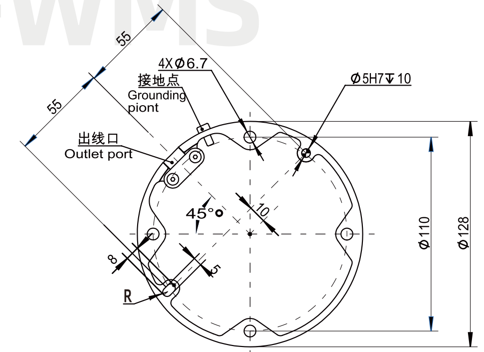
FR3-WML



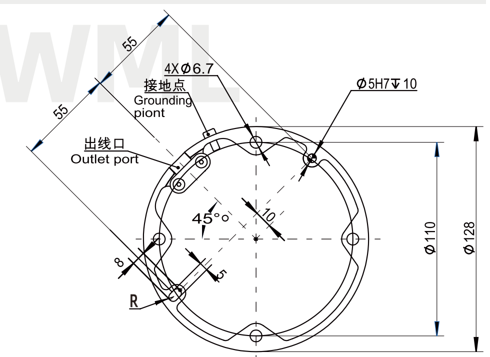
FR3 基座图
Pedestal diagram



FR3-C 基座图
Pedestal diagram



FR3-WMS 基座图
Pedestal diagram

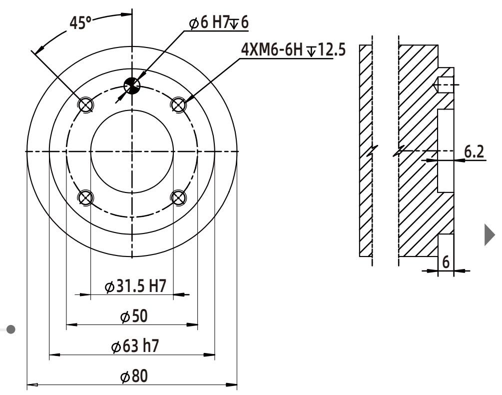


FR3-WML 基座图
Pedestal diagram

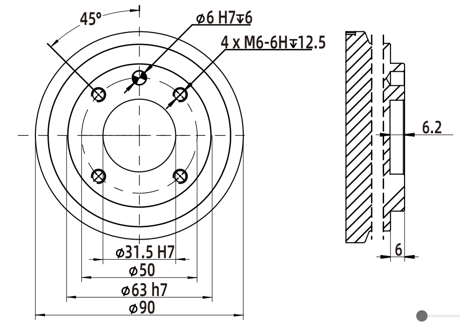
DRAWINGS

技术图纸

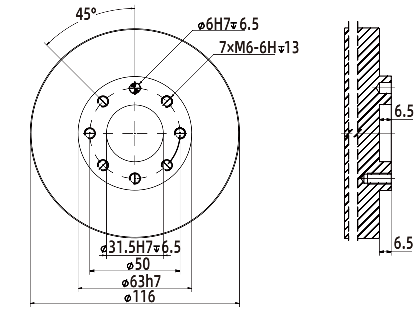
单位(Unit) : mm



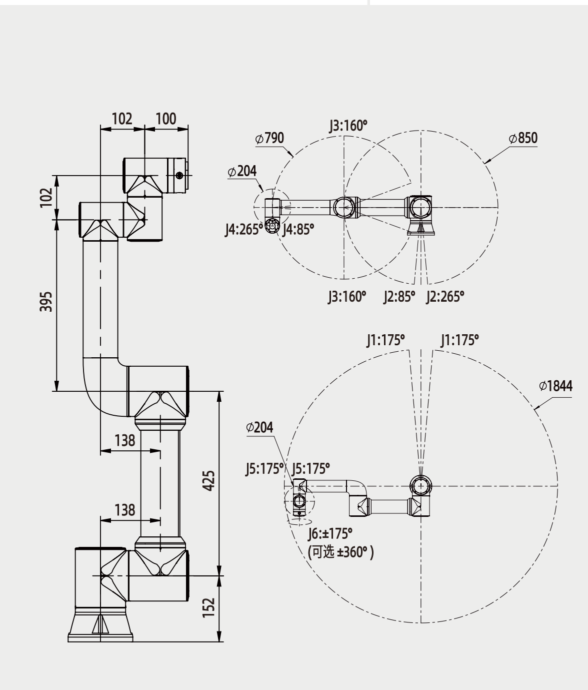
机器人末端
均采用国际标准



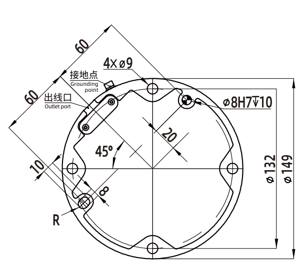
机器人末端
均采用国际标准



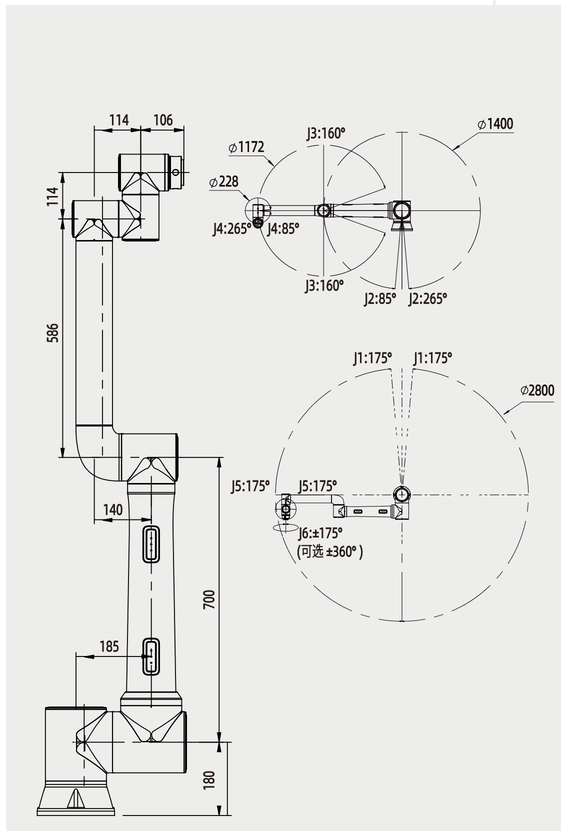
机器人末端兼容
工业机器人末端连接方式



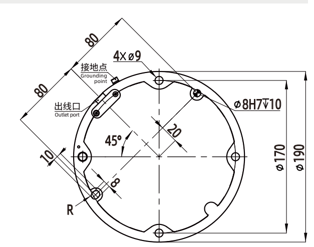
FR5



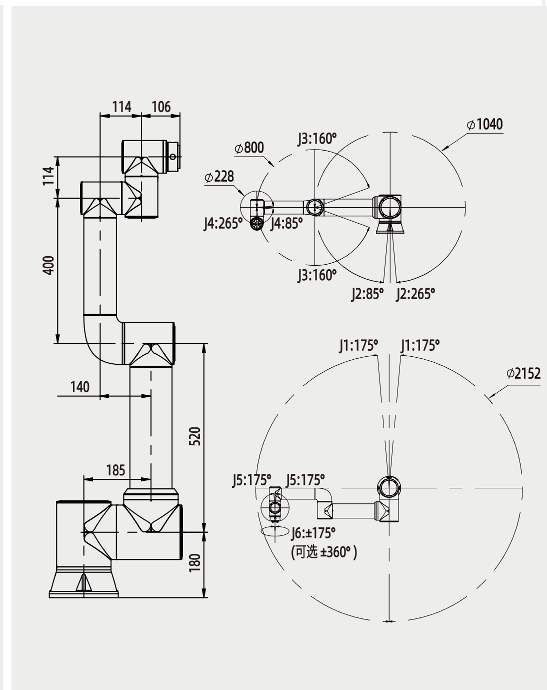
FR5 基座图
Pedestal diagram



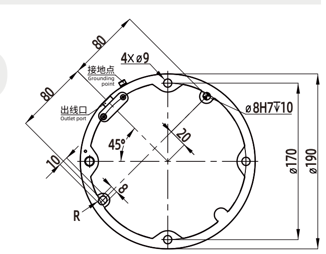
FR10



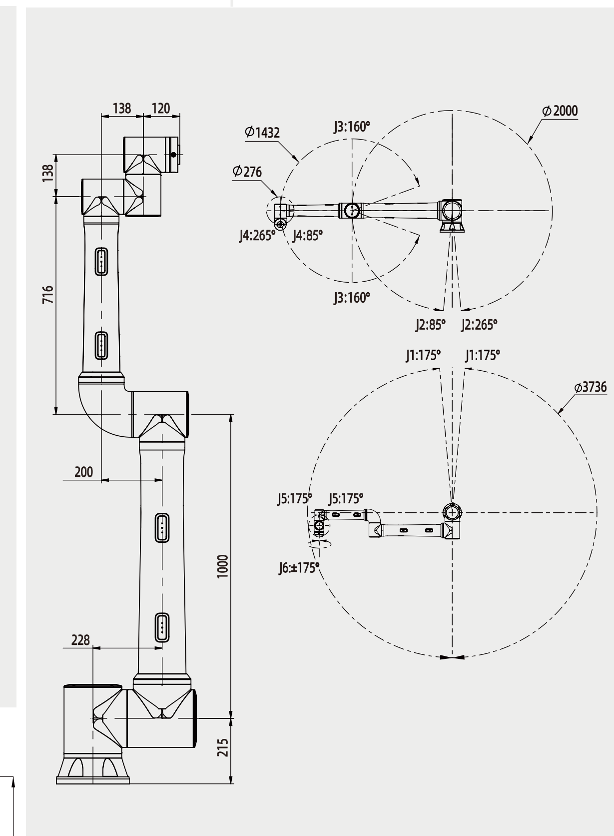
FR10 基座图
Pedestal diagram



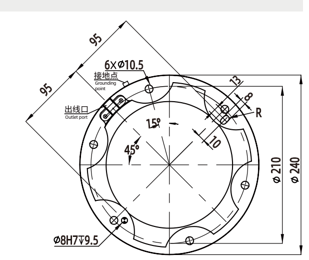
FR16



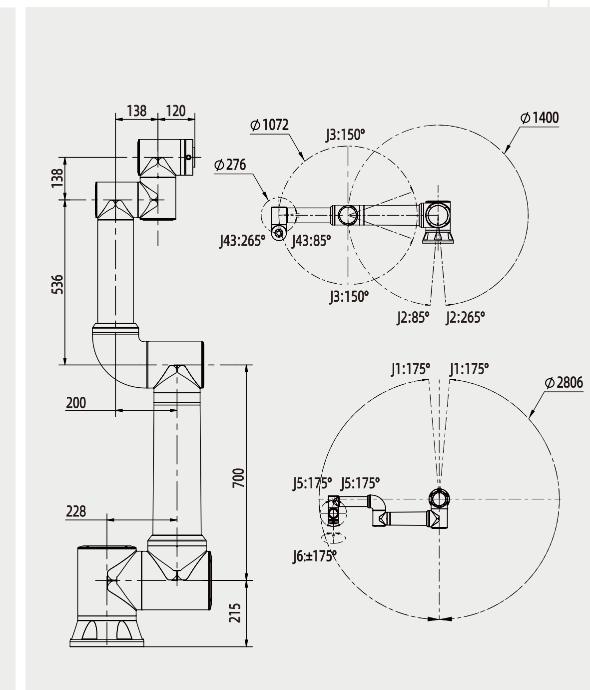
FR16 基座图
Pedestal diagram



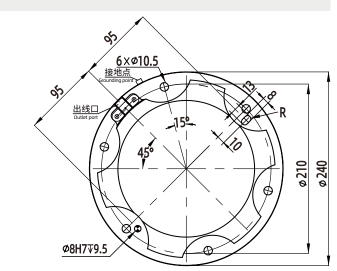
FR20



FR20 基座图
Pedestal diagram



FR30



FR30 基座图
Pedestal diagram